

## WINDHAM, CT, USA

WMO: 725084

Lat: 41.742N

Lon: 72.184W

Elev: 75

StdP: 100.42

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54767

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.1      | -12.8 | -23.4                         | 0.5 | -13.0 | -21.0 | 0.6 | -10.6 | 10.2                  | 0.5  | 9.1 | -1.2 | 2.0                   | 290  | 0.480 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 32.1            | 23.0 | 30.6 | 22.3 | 29.0 | 21.3 | 24.7                | 29.4 | 23.9 | 28.0 | 23.0 | 26.7 | 3.8                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.2                            | 18.1 | 26.6 | 22.5 | 17.4 | 25.9 | 22.0 | 16.9 | 25.4 | 75.1          | 29.4 | 71.7 | 28.1 | 68.6 | 26.8 | 30.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.6           | 7.6  | 6.8 | -19.7                      | 34.9 | 2.9                | 1.5 | -21.8                                              | 36.1 | -23.6      | 37.0 | -25.2      | 37.8 | -27.3      | 39.0 |    |    |
| (5)               |      |     | -20.2                      | 26.4 | 2.8                | 1.2 | -22.2                                              | 27.2 | -23.9      | 27.9 | -25.5      | 28.6 | -27.5      | 29.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.4   | -2.5  | -0.9  | 3.1   | 9.3   | 15.0  | 19.6  | 22.7  | 21.8  | 17.9  | 11.5  | 5.7   | 0.7   |  |  |
|                                                                        | DBStd       | 9.74   | 5.83  | 5.12  | 5.00  | 4.38  | 4.07  | 3.59  | 2.82  | 2.97  | 3.75  | 4.53  | 4.63  | 5.13  |  |  |
|                                                                        | HDD10.0     | 1453   | 387   | 305   | 222   | 63    | 4     | 0     | 0     | 0     | 0     | 37    | 143   | 292   |  |  |
|                                                                        | HDD18.3     | 3279   | 644   | 537   | 473   | 273   | 120   | 28    | 2     | 5     | 52    | 217   | 378   | 548   |  |  |
|                                                                        | CDD10.0     | 1594   | 1     | 1     | 7     | 42    | 160   | 287   | 394   | 364   | 238   | 83    | 15    | 3     |  |  |
|                                                                        | CDD18.3     | 379    | 0     | 0     | 0     | 2     | 17    | 65    | 138   | 112   | 41    | 4     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 3343   | 0     | 0     | 4     | 45    | 232   | 609   | 1203  | 920   | 302   | 27    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 1003   | 0     | 0     | 1     | 14    | 72    | 190   | 396   | 263   | 66    | 2     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.5    | 2.7   | 2.8   | 3.2   | 3.1   | 2.7   | 2.3   | 2.1   | 1.9   | 2.0   | 2.4   | 2.4   | 2.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1220   | 95    | 83    | 107   | 104   | 92    | 104   | 88    | 101   | 113   | 105   | 99    | 117   |  |  |
|                                                                        | PrecMax     | 1508   | 197   | 194   | 272   | 239   | 185   | 281   | 270   | 264   | 242   | 359   | 180   | 217   |  |  |
|                                                                        | PrecMin     | 832    | 45    | 39    | 13    | 31    | 25    | 4     | 37    | 41    | 25    | 15    | 24    | 40    |  |  |
|                                                                        | PrecStd     | 174    | 38    | 33    | 60    | 53    | 35    | 74    | 42    | 57    | 55    | 77    | 39    | 40    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 15.5  | 16.3  | 22.9  | 29.0  | 32.1  | 32.9  | 34.2  | 33.0  | 31.3  | 26.3  | 21.0  | 17.7  |  |  |
|                                                                        |             | MCWB   | 13.2  | 11.9  | 15.1  | 17.9  | 20.3  | 23.2  | 24.4  | 24.2  | 22.9  | 19.1  | 15.3  | 14.2  |  |  |
|                                                                        | 2%          | DB     | 12.2  | 12.7  | 17.5  | 23.3  | 28.6  | 31.0  | 32.2  | 31.1  | 28.4  | 23.5  | 18.0  | 14.3  |  |  |
|                                                                        |             | MCWB   | 9.8   | 9.6   | 10.9  | 13.5  | 19.0  | 22.2  | 23.6  | 23.0  | 20.8  | 18.4  | 14.3  | 12.4  |  |  |
|                                                                        | 5%          | DB     | 8.7   | 9.8   | 13.8  | 20.2  | 26.0  | 28.9  | 30.6  | 29.3  | 26.8  | 21.5  | 16.3  | 11.8  |  |  |
|                                                                        |             | MCWB   | 6.4   | 6.3   | 9.0   | 12.2  | 17.4  | 20.9  | 22.7  | 21.9  | 20.2  | 17.4  | 12.7  | 9.9   |  |  |
|                                                                        | 10%         | DB     | 5.9   | 7.1   | 11.3  | 17.5  | 23.0  | 27.2  | 28.9  | 27.9  | 24.8  | 19.2  | 14.0  | 8.8   |  |  |
|                                                                        |             | MCWB   | 3.3   | 3.9   | 7.3   | 10.8  | 15.9  | 20.0  | 21.7  | 21.2  | 19.3  | 15.7  | 10.7  | 6.3   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 13.8  | 13.7  | 15.5  | 18.5  | 22.2  | 24.8  | 26.0  | 25.9  | 24.2  | 21.9  | 17.6  | 15.5  |  |  |
|                                                                        |             | MCDB   | 14.9  | 15.4  | 21.4  | 26.5  | 29.2  | 30.8  | 32.0  | 30.9  | 28.0  | 23.7  | 19.3  | 17.2  |  |  |
|                                                                        | 2%          | WB     | 10.3  | 10.0  | 12.4  | 16.3  | 20.5  | 23.6  | 24.8  | 24.6  | 22.9  | 20.0  | 15.6  | 12.6  |  |  |
|                                                                        |             | MCDB   | 11.5  | 12.0  | 15.5  | 21.1  | 25.9  | 28.8  | 30.0  | 28.4  | 26.0  | 22.3  | 17.5  | 14.2  |  |  |
|                                                                        | 5%          | WB     | 6.5   | 6.5   | 10.1  | 14.3  | 19.0  | 22.4  | 24.0  | 23.7  | 22.1  | 18.1  | 13.7  | 9.8   |  |  |
|                                                                        |             | MCDB   | 8.3   | 9.1   | 13.3  | 18.0  | 24.0  | 26.8  | 28.3  | 27.2  | 24.9  | 20.5  | 15.7  | 11.4  |  |  |
|                                                                        | 10%         | WB     | 3.5   | 4.1   | 7.6   | 12.2  | 17.5  | 21.2  | 23.1  | 22.8  | 21.0  | 16.3  | 11.3  | 6.6   |  |  |
|                                                                        |             | MCDB   | 5.3   | 6.6   | 10.7  | 15.9  | 21.1  | 24.9  | 26.8  | 26.1  | 23.6  | 18.9  | 13.5  | 8.5   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.5   | 10.2  | 10.6  | 11.9  | 12.0  | 11.2  | 11.1  | 10.9  | 11.6  | 11.3  | 10.6  | 9.3   |  |  |
|                                                                        |             | MCDBR  | 12.5  | 13.9  | 15.1  | 16.9  | 16.4  | 14.0  | 12.6  | 12.1  | 13.0  | 13.0  | 13.2  | 12.5  |  |  |
|                                                                        |             | MCWBR  | 10.7  | 10.7  | 9.7   | 8.7   | 7.7   | 6.6   | 5.4   | 5.1   | 6.6   | 8.1   | 9.8   | 11.2  |  |  |
|                                                                        |             | MCDBR  | 12.3  | 13.0  | 13.4  | 13.5  | 13.5  | 12.0  | 10.5  | 9.7   | 9.9   | 10.7  | 11.4  | 11.8  |  |  |
|                                                                        | 5% WB       | MCWBR  | 11.3  | 11.3  | 10.1  | 8.4   | 7.2   | 6.3   | 5.0   | 4.8   | 6.0   | 8.1   | 9.9   | 11.3  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.295 | 0.304 | 0.328 | 0.370 | 0.414 | 0.457 | 0.476 | 0.443 | 0.372 | 0.335 | 0.308 | 0.294 |  |  |
|                                                                        | taud        |        | 2.479 | 2.465 | 2.462 | 2.373 | 2.279 | 2.195 | 2.150 | 2.231 | 2.408 | 2.526 | 2.559 | 2.554 |  |  |
|                                                                        | Ebn at Noon |        | 862   | 908   | 923   | 900   | 864   | 824   | 802   | 819   | 861   | 861   | 842   | 836   |  |  |
|                                                                        | Edh at Noon |        | 74    | 88    | 99    | 117   | 132   | 143   | 149   | 133   | 104   | 82    | 69    | 64    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.85  | 2.71  | 3.71  | 4.79  | 5.32  | 5.78  | 5.95  | 5.32  | 4.36  | 2.90  | 2.02  | 1.53  |  |  |
|                                                                        | RadStd      |        | 0.14  | 0.24  | 0.27  | 0.52  | 0.55  | 0.58  | 0.33  | 0.28  | 0.33  | 0.28  | 0.16  | 0.15  |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +107    | N/A     |
| (47) Regional (81 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +98     | +50     |

Nomenclature: See separate page